Def.

A deformation retraction of a space X onto a subspace A is a family

$\{ f_t : X \to X \mid t \in I \}$ satisfying $f_0 \equiv id$, $f_1[X] = A$, $f_t|_A \equiv id_A$ for all t,

and $X \times I \to X : (x,t) \mapsto f_t(x)$ is continuous.

More generally, define

Def.

A Homotopy of a space X into a space Y is a family

$\{ f_t : X \to Y \mid t \in I \}$ satisfying the associated map $F : X \times I \to Y : (x,t) \mapsto f_t(x)$ is continuous.

Moreover, given two maps g_0, g_1 are called Homotopic if there is a homotopy.
In such a situation, denote $g_0 \simeq g_1$

Recall that: a map $r : X \to X$ satisfying $r[X] = A$ and $r|_A \equiv id_A$ is called retraction.
Thus, we can say that the deformation retraction of a space X onto a subspace A is a homotopy from the identity map of X to a retraction of X onto A.
And, algebraically, the retraction $r : X \to X$ satisfies that $r \circ r = r^2 = r$, analogous to the projection operator.

Def.

A map $f : X \to Y$ is called a homotopy equivalence if: there exists a map $g : Y \to X$ s.t.

$$f \circ g \simeq id_Y \quad \text{and} \quad g \circ f \simeq id_X.$$

If such a map exists, then the spaces X and Y are called homotopy equivalent.

Def.

A space that homotopy equivalent to a point is called Contractible.